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An Intelligent Traffic Management System with Adaptive Traffic Light Control Based on Vehicle Density Estimation

1st Gopi Sri Harsha Karedla
Department of Computer Science and Engineering
Velagapudi Ramakrishna Siddhartha Engineering College
Vijayawada, India
gopisriharsha4566@gmail.com

2nd Ashutosh Satapathy*
Department of Computer Science and Engineering
Velagapudi Ramakrishna Siddhartha Engineering College
Vijayawada, India
ashutosh@vrsiddhartha.ac.in

3rd Altaf Hussain Shaik
Department of Computer Science and Engineering
Velagapudi Ramakrishna Siddhartha Engineering College
Vijayawada, India
218w1a05b6@vrsec.ac.in

4th Sai Sowrabh Nallagondla
Department of Computer Science and Engineering
Velagapudi Ramakrishna Siddhartha Engineering College
Vijayawada, India
218w1a05a4@vrsec.ac.in

***Abstract*—The Intelligent Traffic Management System utilizes computer vision and advanced traffic control techniques to detect two- and four-wheelers precisely and estimate vehicle density at intersections. Urban areas becoming increasingly congested, so efficient traffic control is essential. Traditional traffic lights operate on fixed intervals, often causing delays and long waits during peak hours. This proposed system utilizes Faster R-CNN accurate vehicle detection, while a mixture of Gaussian model aids in vehicle density estimation. Additionally, fuzzy logic is implemented to adaptively control traffic light timings based on real-time data, prioritizing high-traffic flows to reduce congestion and travel times. This approach has shown promise when integrated with existing frameworks of managing traffic which can be expanded as needed; it has been found to have an accuracy level of 93.2%, a precision rate of 95.4%, and a recall rate of 91%, demonstrating its effectiveness and adaptability**

Keywords—intelligent traffic management, congestion reduction, vehicle detection, vehicle density analysis, traffic light control, computer vision.

I. Introduction

Recent statistics from Statista reveal that the average congestion index in the target city has risen by a record 11.3% over the past five years, with 7% more accidents recorded. Intelligent Traffic Management System (ITMS) is deployed to manage traffic effectively during heavy congestion as a result of this. Typically, traffic lights operate on fixed intervals for all directions, which can be inefficient for those stuck in heavy traffic. Prior research has used Artificial Neural Networks (ANNs) [1], particularly deep neural networks, which are computationally intensive and time-consuming. This can affect the calculation of performance metrics. Additionally, many models rely on sensors to collect data on vehicles, including license plate numbers, which raises privacy concerns and requires regular maintenance to ensure accuracy and reliability. Licensing fees for software components, ongoing maintenance costs, and the need for specialized personnel can also contribute to higher expenses [2]. This research introduces an adaptive traffic light control system that utilizes real-time vehicle detection to adjust signal timings based on vehicle density, thereby reducing congestion and wait times. By leveraging advanced vehicle density estimation and dynamic signal adaptation, it enhances traffic flow and intersection efficiency in urban environments. The proposed system uses computer vision and convolutional neural network algorithms to analyze real-time video feeds from strategically placed cameras at intersections. Computer vision technology is crucial in ITMS, particularly in vehicle density analysis [3]. This paper presents new algorithms, methodologies, and techniques for vehicle tracking and computer vision-based traffic surveillance. Fig. 1 provides an illustration of the vehicle detection process which is the first stage of this work. However, Precise vehicle detection is essential for identifying vehicles in congested areas, minimizing gridlocks, and making travel easier during peak hours. ITMS combines various components into a coherent and adaptive traffic management infrastructure.

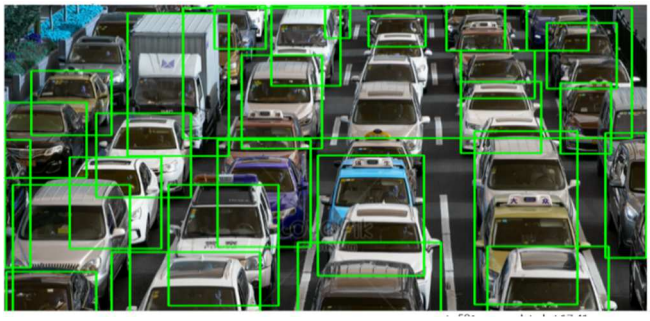


Fig. 1. Vehicle detection using deep learning model

Vehicle detection is vital for recognizing vehicles in traffic areas, which helps reduce congestion effectively and improve travel efficiency at peak hours. This detection is performed using Faster Region-based Convolutional Neural Network (FR-CNN), which provides accurate and efficient identification of vehicles in complex traffic scenarios Fig .1. Traffic density estimation refers to the process of quantifying either the number of cars on certain road sections or congestion levels in defined regions over a specified period. To enhance computer vision methods used for traffic analysis [4], this article presents an original technique that uses Gaussian Mixture Model (GMM) to estimate accurately vehicular densities from video data. In other words, what this means is that through such like methods, it becomes possible for managers of transportation systems to adjust signal timings dynamically depending on prevailing real-time vehicle counts along different parts of roads so as to reconfigure lanes or even put in place enforcement measures where necessary thus

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greatly reducing traffic jams while at the same time increasing general flow of traffic.

Adaptive Traffic Light Control is a system that adjusts traffic signal timings based on actual traffic demand, using Adaptive Signal Control Technologies (ASCT) [5]. This method uses fuzzy logic algorithm, a more stable alternative to fixed-time traffic signal systems, to adjust green light timings based on real-time situations [6]. This system automatically responds to unexpected vehicle movements over intersections, reducing congestion and reducing negative effects like wasted time, increased air pollution, and higher economic costs. Adaptive Traffic Light Control synchronizes green lights, facilitating smooth vehicular flow and preventing stop-and-go movements, ultimately contributing to a more efficient and environmentally friendly transportation system. The aim is to reduce travel times by synchronizing green lights, preventing stop-and-go movements, and contributing to a more efficient and environmentally friendly transportation system. This article reviews traffic management systems, proposes a methodology for vehicle detection, density estimation, and adaptive traffic light control, discusses experimental setup and results, and presents conclusions and future work.

II. Related work

Deshpande and Hsieh used Convolutional Neural Networks (CNNs), ANNs, and Support Vector Machine (SVMs) to estimate traffic conditions and optimize traffic light timings [7]. The vehicle detection system was highly accurate with DarkNet-53 having an accuracy of 97.5% and SVM achieving validation accuracy of 98.191% in predicting the green light timing control. The challenges of this article include running a complex cyber-physical system can consume a significant amount of energy, which may not align with sustainability goals. On the other hand, Bisht et al. applied You Only Live Once (YOLO) for traffic congestion analysis as well as Single Shot Detector (SSD) and Region-based CNN (RCNN) algorithms to calculate vehicle densities [8]. The algorithms were used to compute green phase durations based on lane numbers and maximum Passenger Car Unit (PCU) counts. The research demonstrated average mean precisions of 63.2% using YOLOv3, 69.8% using SSD and Mask-RCNN which had an average mean precision of 72.4% for object detection. Local traffic dynamics can be complex and unpredictable. An indigenous system might struggle to handle unique traffic scenarios that were not accounted for during its design.

Abdullah et al. employed Random Forest algorithm and trained using past traffic flow patterns, historical weather conditions and other pertinent factors to predict future traffic in real-time [9]. Additionally, the researchers used recurrent neural networks for different applications of Intelligent Transportation Systems (ITS) including traffic forecasting, detection of jams and optimization of traffic lights. The study drew its data from INRIX and Caltrans Performance Measurement System (PeMS), which feature both real-time and historical information on road use obtained through GPS-enabled gadgets like mobile phones and navigation appliances. Implementation of advanced traffic management systems and infrastructure can be expensive, requiring significant financial investments. Hence, this approach managed to resolve congestion with an accuracy rate of 87.5% and precision level at 94.36%. Saleem et al. applied Fusion-based intelligent traffic congestion control system for VNs (FITCCS-VN) that collect traffic data and route traffic on available route, Random Forest Classification algorithm to construct a model for traffic congestion state perception and Supporting Vector Machine to perceive a hyperplane that can disperse the data precisely, in addition to the best need to be found [10]. This paper had considered prelabelled VN dataset which contains 2282 instances and 515 features, of which 514 attributes are independent and one, the output class, is dependent. This paper obtained 95% accuracy and 62.5% precision. Implementation of a complex traffic control system may require significant energy consumption and contribute to environmental issues.

The smart traffic light algorithm introduced by Alsaawy et al. is one of the many focused on congestion management in an intersection [11]. The research used second level image processing algorithms for decision making and incorporating RFID for object identification and WSNs to collect environmental information. This system had a very impressive 95 % vehicle detection rate. These systems are typically deployed in specific areas, which means they may not provide a comprehensive view of traffic conditions across an entire city or region. Gaps in coverage can limit their effectiveness. Ali et al. utilized an adaptive traffic control approach incorporating fuzzy logic, optimized with Webster's and modified Webster's algorithms, to enhance traffic flow at intersections [12]. The research involved simulations with generated and real field-collected data for three- and five-approach intersections, comparing adaptive methods against fixed-time control strategies. Results indicated improvements in average travel time, speed, and waiting time for four-approach intersections, but a decrease in average speed at five-approach intersections due to frequent phase switching. The adaptive fuzzy logic method achieved a 90.325% accuracy in pixel data, with an IoU of 67.543% in predicting the green light timing control.

Neelakandan et al. proposed the Optimized Weight Elman Neural Network (OWENN) algorithm to enhance traffic management using the Intel 80,286 Microprocessor. The system involves five phases: Internet of Things (IoT) data collection, feature extraction, classification, optimized traffic IoT values, and traffic signal control [13]. This paper collected IoT traffic data from a kaggle dataset. The OWENN model achieved 98.23% accuracy and 96.69% f-measure, demonstrating superior efficiency. Mathiane et al. employed 2D Convolutional Neural Network (2D-CNN) for vehicle classification and signal switching algorithm for adaptive traffic light control. This paper considered data collected from photos of various vehicles including cars, trucks, buses, and ambulances, is divided into training datasets and test datasets [14]. This paper had achieved 96% accuracy in classifying vehicles using the test dataset. The study relies heavily on simulations in controlled environments using Pygame and SUMO to validate the system. This may not fully represent real-world complexities such as diverse traffic patterns and unpredictable events.

Yang et al. proposed a multi-object detection algorithm using DarkNet-53 and dense convolution network (DenseNet) [15]. This article conducted experiments on the public KITTI dataset which contains 7481 stereo images, including 8 different categories: car, truck, van, pedestrian, sitting people, cyclist, tram and mixture and Pascal VOC datasets which includes 11,540 images and 20 categories that are common in everyday life. This paper had achieved 88.84% mAP on

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Pascal VOC and 91.13% mAP on KITTI datasets. Both DarkNet-53 and DenseNet have high computational requirements due to their deep architectures, which can lead to increased latency. Reddy et al. explored the architectural intricacies and methodological differences of three CNN-based object detection algorithms, namely FR-CNN, YOLOv3, and SSD in the specific domain application of vehicle detection [16]. This paper had considered two datasets namely Caltech Cars 1999 which contained 126 images of cars with a size of 896 × 592 pixels and Caltech Cars 2001 which contained 526 images of cars with a size of 360 × 240 pixels. The FR-CNN algorithm achieved an accuracy of 99.58% with a mean average precision (mAP) of 76%. YOLO v3 reached an accuracy of 99.47% and a mAP of 81%, while SSD achieved 99.36% accuracy with a higher mAP of 92%.

III. Proposed Methodology

Traffic congestion and accident risks will be minimized, optimizing traffic flow and reducing emissions by Intelligent Traffic Management System which enhances road safety. It also promotes smart city solutions using real-time data on vehicle density for adaptive control and urban planning.

A. Process flow Diagram

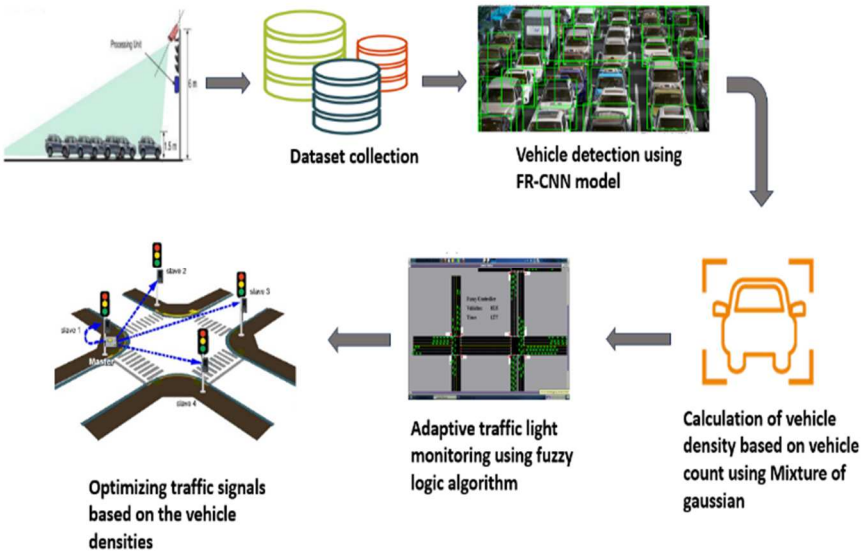


Fig. 2. Pipeline architecture for intelligent traffic management system

Fig. 2 portrays the proposed system architecture together with its methodology modules for a computer vision-based traffic light management prototype. Starting off with hardware configuration, dataset collection, utilization of a Faster R-CNN model for vehicle detection, calculating density. Furthermore, Adaptive Traffic Light Control is a combination of fuzzy logic algorithms driving it intelligently manages the phases of traffic based on live vehicle densities to ensure efficiency in an ever-changing environment.

The proposed density-based traffic light management system initially leverages online dataset for traffic surveillance, an efficient processing unit and a consistent network. This system employs Faster RCNN (FRCNN) model for vehicle detection by quickly generating region proposals for target objects detection. This model effectively detects vehicles of various sizes and positions, including distant and partially occluded ones, by using a region proposal network to isolate and process objects at different scales. A stochastic technique known as Mixture of the Gaussian (MoG) is used for removing background in video processing. It is the statistical approach that utilizes the idea of a mixture of Gaussian backgrounds for each pixel intensity within the background. This makes it possible to separate moving cars with that of the fixed background thus enabling effective vehicle and density measurement. Fuzzy Logic algorithms which consider multiple variables such as density of traffic in making decisions concerning the traffic light system are employed in traffic state analysis. The type of system used in this research offers traffic optimizations which eliminate the causes of delays so that an adequate traffic flow is provided.

B. Model Selection

The choice of FRCNN model in ITMS exhibit several benefits, making it a right option for vehicle identification and tracking in dynamic traffic scenarios [17]. This model is known for its highest level of precision in object detection tasks. It captures complex features as well as spatial relationships within the traffic scenes to enable reliable identification of vehicles, pedestrians, among others. Faster R-CNN effectively detects vehicles at various sizes and positions using multi-scale bounding boxes, though small or occluded vehicles may slightly reduce accuracy.

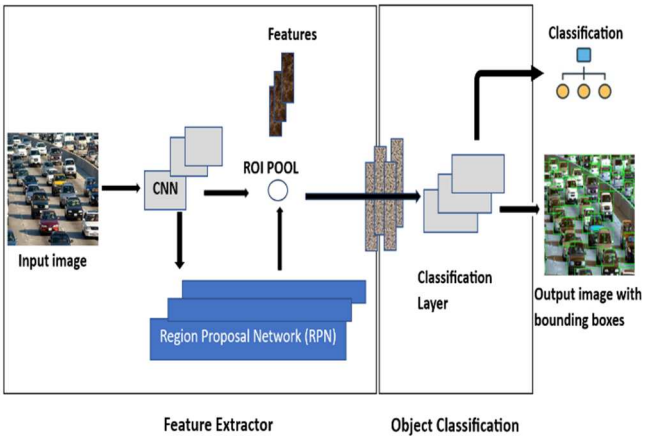


Fig. 3. Architecture of FR-CNN model for vehicle detection

This specific section aims at shedding light on the FR-CNN architecture used for object detection as shown in Fig.

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3. The input image is first put into convolutional layers, which extract feature maps. Then comes generating region proposals that help in spotting where objects might be found. Region of Interest (RoI) pooling turns those proposals into fixed-size feature maps. Lastly, to do classification and bounding box regression fully connected layers process these feature maps. On one hand, it allows accurate object detection by concentrating on areas of interest and refining boundaries. However, every proposal must be processed individually, thus this could interfere with its performance making it slow and computationally intensive. This article examines how the FRCNN model can be applied in an Intelligent Traffic Management System for vehicle detection and tracking in a dynamic traffic environment due to several advantages it has over other models. It can capture fine-grained details and spatial relationships among objects in traffic scenes hence enabling reliable recognition of vehicles, pedestrians etc.

Algorithm 1: Vehicle density estimation using GMM.

Input: An input image (I) consisting of all types of vehicles.
Output: Total count of vehicles (V_{count}) and density of vehicles (D).

1. Initialize the background subtractor:
   Let *L(t)* and *B(t)* represent the current video frame, and background model at time *t*, initialized using a GMM.
2. Apply the background subtractor to the frame:
   Compute foreground extraction *F(t)* as:
   $$F(t) = L(t) - B(t)$$
   */*Detect moving objects (vehicles) by subtracting background model from the current frame. */*
3. Remove shadows:
   Make changes to foreground mask to remove shadows:
   $$F'(t) = \begin{cases} F(t) & if\ F(t) > Shadow\ Threshold \\ 0 & Otherwise \end{cases}$$
   */*Shadow threshold helps in distinguishing shadows from real objects (usually slightly below 255). */*
4. Find contours in the foreground mask:
   Obtain foreground mask contours:
   $$Contours = \{C_1, C_2, C_3, \ldots C_n\}$$
   */*Each contour C_i represents a possible vehicle or object on foreground mask. */*
5. Filter and count valid contours as vehicles:
   A_{C_i} is area of contour C_i. If $A_{C_i} > A_{min}$, contour contains a vehicle
   $$V_{count} = \sum_{i=1}^{n} 1(A_{C_i} > A_{min})$$
   */*1(·) is the indicator function, which is set at 1 for true conditions and 0 otherwise. */*
6. Compute vehicle density (D):
   Determine frame area A_{frame} and vehicle density *D*:
   $$A_{frame} = WH \text{ and } D = \frac{V_{count}}{A_{frame}}$$
   */*W and H represent width and height of the frame, respectively. */*
7. Return total count of vehicles V_{count} and density of vehicles *D*.

The selection of GMM for vehicle density estimation because it effectively captures variations in background and foreground objects in video feeds, making it highly suitable for accurately modelling dynamic traffic scenes [18]. GMM adapts to gradual lighting changes but may struggle with sudden shifts or extreme weather, requiring additional filtering for best results. The background subtractor is initialized by Algorithm 1, employing a GMM, where L(t) represents the current video frame and B(t) the background model. The foreground extraction F(t) is computed by removing the background model from the current frame. To remove shadows, $F'(t)$ retains the value of F(t) if its pixel value exceeds some shadow suppression threshold; otherwise, it is set to zero. Potential vehicles are represented by contours in the foreground mask and valid contours are those whose area A_{C_i} exceeds a minimum threshold A_{min} resulting in vehicle count V_{count}. D is thus calculated as dividing V_{count} by the frame area A_{frame} that depends on frame width and height.

Algorithm 2: Fuzzy logic for adaptive traffic light control

Input: Traffic light state, phase durations, and traffic density along four lanes.
Output: Traffic light states based on vehicle densities along four lanes.

1. Define Objective Function:
   $F_i(t)$: Traffic flow rate for lane i (North, South, East, West) at time *t*.
   $W_i(t)$: Waiting time of vehicles on lane *i* at time t.
   $W_{total}(t)$: Sum of waiting times in all lanes.
   $$W_{total}(t) = \sum_{i=1}^{4} F_i(t)\, W_i(t)$$
2. Constraints:
   G_{Max}: Maximum green phase duration for any lane.
   G_{Min}: Minimum green phase time for any lane.
   $$G_{Min}(t) \le Green_i(t) \le G_{Max}(t)$$
   C: Cycle time to complete one full cycle of all lights, and it is expressed as
   $$C = \sum_{i=1}^{4} (Green_i(t) + Yellow_i(t) + Red_i(t))$$
3. Define the fuzzy rules:
   $\mu_{flow_low}(x)$: An extent to which a given traffic density *x* falls under *low* category.
   $$\mu_{flow_low}(x) = \begin{cases} \frac{\max(0, 1-(x-f_{low}))}{(f_{med}-f_{low})}, & if\ x < f_{med} \\ 0, & otherwise \end{cases}$$
   f_{low} and f_{med} are predetermined medium and low traffic density.
4. Fuzzy Implication:
   $I(A,B)(x)$: Fuzzy rule represents relationship between two fuzzy sets A and B.
   $$I(A,B)(x) = \min(\mu_A(x), \mu_B(x))$$
   */*Models different traffic situations to determine the green light condition*/*
5. Aggregate rule outputs:
   w_i: Assigned weight to i^{th} fuzzy rule.
   $Green_output$: Output represents combined green light time based on all fuzzy rules.
   $$Green_output = \frac{\sum_i w_i \times Greenrule_i(t)}{\sum_i w_i}$$
6. Defuzzify outputs:
   G_j : Possible values of green light duration allocated to lane *i*.
   $u_{aggregated_output_i}(G_j)$: Aggregated membership function for lane *i* at duration G_j.
   $Green_Duration_i$: Defuzzified green light duration for lane *i*
   $$Green_Duration_i = \frac{\sum_j G_j \times u_{aggregated_output_i}(G_j)}{\sum_j u_{aggregated_output_i}(G_j)}$$
7. Update traffic light timings:

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Set traffic light to computed green light duration for each lane

$$Green_i(t) = Green_Duration_i$$

The decision to use fuzzy logic for adaptive traffic light control because it allows for flexible, real-time adjustments to signal timings based on traffic flow, providing a more responsive approach than fixed-time systems in varying traffic conditions [5]. Green light durations are initially based on typical traffic flow but are adjusted in real-time according to current traffic density to optimize flow and reduce congestion. Algorithm 2 exhibits the traffic light control methodology that is based on fuzzy logic that allows for the adjustment of the green light period in relation to real-time traffic at the four-way intersections. It takes as inputs the traffic flow rate $F_i(t)$ and waiting time $W_i(t)$ for each lane i (North, South, East and West). The system sums the waiting time over all the lanes $W_{total}(t)$ and determines light periods $G_i(t)$ within restrictions as maximum period G_{max}, minimum period G_{min}, and overall cycle time C. The fuzzy inference processes include setting such conditions as low traffic flow, or high waiting time, and assigning membership to these conditions and using fuzzy rules to define the length of the green light. The green light period at the end of the cycle for each direction is further defuzzified using a form of weighted mean considering the potential waiting times G_j and their respective membership weight which leads to improved overall transition and waiting time performance of the traffic signals.

IV. Results and Discussion

A. Experimental Setup

Vehicle detection and density estimation require extensive data due to diverse traffic scenarios at intersections. To build the dataset, images were collected from Google and iStock videos under authorized licensing agreements and tagged manually or using pretrained models, with COCO dataset images added for object detection consistency [19]. Data cleaning is done by removing duplicates, resizing, and labeling into training, testing datasets, with manual annotation where needed. Data augmentation was applied to increase sample diversity. System requirements for this model include a modern OS (Windows 7 or 10/Mac OS X 10.11 or higher), x86 64-bit CPU, 4GB SSD, and 4GB DDR4 RAM. Software requirements include: NumPy for array manipulation, Google Colab for a collaborative machine learning environment, TensorFlow for model development, OpenCV for computer vision tasks, Matplotlib for data visualization, and Scikit-Fuzzy for fuzzy logic operations.

B. Performance Metrics

True Positive (TP) is defined as the model's performance in the identification of the vehicles. Equation (1) calculates the true positives, which describes the number of vehicles that have been identified as present in the image or video frame by the model.

$$TP = \sum_{i=1}^{N} 1(IoU(b_i^{pred}, b_i^{gt})) \geq \tau \quad (1)$$

N corresponds to all the objects that have been detected. b_i^{pred} is the bounding box obtained from the predicted label for ith object detected. b_i^{gt} is the ground truth bounding box of the ith object. IoU is defined as the Intersection over Union of the predicted and the true bounding boxes for the object. τ is the threshold of IoU, usually the value is kept at 0.5 for most of the applications. False Positive (FP) represents the false detections made by the model which are not vehicles. The formula for false positive is depicted in Equation (2). It is very important to concentrate on reducing FP to prevent any misunderstandings that may result in false interpretations in the vehicle density analysis.

$$FP = N_{detected} - TP \quad (2)$$

False Negative (FN) is the number of actual vehicles that the model has failed to detect. Equation (3) gives the formulation for false negatives. High FN values imply that vehicles were present but were somehow not accounted for, leading to gross underestimation of vehicle density.

$$FN = N_{groundtruth} - TP \quad (3)$$

Equation (4) refers to accuracy which is an overall model performance metric defined as the percentage of instances predicted correctly as compared to the total number of instances [20]. It is regarding the proportion of instances accurately represented. A high accuracy score indicates that the model is credible and can differentiate vehicles from background regions enhancing the confidence in the analysis on the density of vehicles.

$$Accuracy = \frac{TP+TN}{TP+FP+TN+FN} \quad (4)$$

Precision is defined as the number of true positive predictions made relative to the total number of positive predictions made; it helps determine the accuracy of the positive class only [21]. The formula for precision is presented in Equation (5). High precision translates to a minimal FP, which is paramount when considering the total number of vehicles on a lane.

$$Precision = \frac{TP}{TP+FP} \quad (5)$$

Recall measures the number of correctly predicted positive observations against the total number of actual positive observations [21]. The formula for recall, presented in Equation (6), helps us understand how many true positives the model identifies. It is a crucial parameter that ensures effective vehicle detection by capturing a significant number of vehicles.

$$Recall = \frac{TP}{TP+FN} \quad (6)$$

Mean Squared Error (MSE) is defined as the average of all squared differences between the actual (y_i) and expected outcomes (y_i') in relation with the model to ascertain its accuracy in predicting. The lower the values of MSE, the better the model performance. The formula for MSE is provided in Equation (7).

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - y_i')^2 \quad (7)$$

Root Mean Squared Error (RMSE) is derived by taking the square root of MSE, giving an error interpretation in the target variable's unit. RMSE is one of the simple methods that gives

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an estimate of accuracy of predicted values by the model. The formula for RMSE is shown in Equation (8).

$$RMSE = \sqrt{MSE} \tag{8}$$

Equation (9) represents Mean Absolute Error (MAE) which assesses the average of absolute differences between actual values (y_i) and their predicted values (y_i') regardless of the significance of differences in prediction.

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - y_i'| \tag{9}$$

TABLE I. PERFORMANCE METRICS OF VEHICLE DETECTION

Performance Metrics	Value
Accuracy	95.4%
Precision	93.2%
Recall	91%

TABLE II. PERFORMANCE METRICS OF VEHICLE DENSITY ESTIMATION

Performance Metrics	Value
MSE	4.0%
RMSE	2.0%
MAE	2.0%

Table I present the results concerning the performance evaluation of the FRCNN model for vehicle detection. The model produces an accuracy of 95.4%, precision of 93.2%, and recall of 91%, meaning it captures all vehicles in an image or video. Table II shows the GMM algorithm in vehicle density estimation obtains an MSE of 4.0%, which means that the average error between the predictions and the actual values is very low. Also, RMSE is at 2.0%, and MAE is at 2.0%, which reflects a lower average magnitude of errors in a set of predictions.

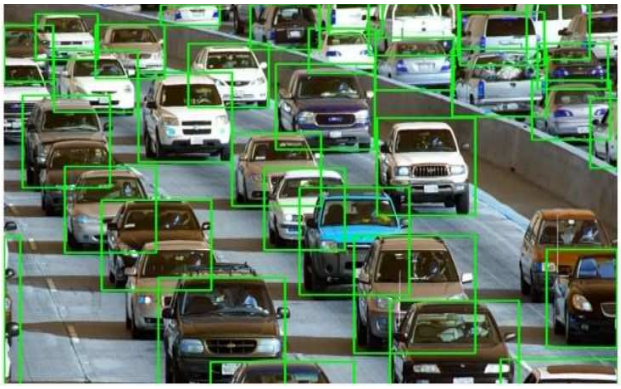


Fig. 4. Vehicle Detection using FRCNN

Fig. 4 demonstrates the bounding box detection, which is necessary to undertake at the early steps of the implementation. The iterative process propelled by bounding box output validation and optimization enabled us to gradually adjust the detectors to enhance the system's performance over time. The desired implementation of the intelligent traffic management system is realized in the real-time optimization of traffic signals based on the information obtained from the vehicle density analysis. This output is vital as the system detects and traces various objects, specifically vehicles, through the processing of the video frames and provides an output of the present vehicle population on busy roads. The fuzzy logic algorithm utilizes this information.

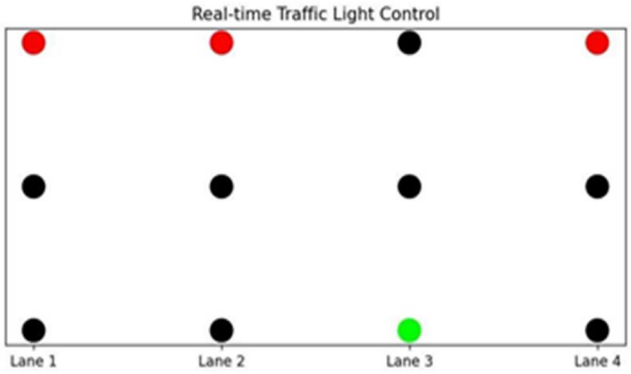


Fig. 5. Adaptive traffic signal display

Smart traffic lights tend to control adaptive light management, the output of which is illustrated in Fig. 5. Traffic control measures which include the continual adjustments of traffic lights to the prevailing situation in terms of vehicle density. This allows current vehicle density to be yielded as the system detects and follows vehicles in processed video frames. This information becomes an important input in fuzzy logic control, whose output is further processed with the intelligent logic of appropriate changes in traffic light timings for an effective traffic control system.

TABLE III. VALUES OF ADAPTIVE TRAFFIC LIGHT CONTROL

Lane No.	Density	Duration (sec)
Lane 1	36	33.540372692
Lane 2	25	23.291925463
Lane 3	60	55.900621138
Lane 4	40	37.267080774

As seen in Table III, the values last depicted pertain to the duration of time spent in each lane. The values are derived from the vehicle densities in each of the lanes. For example, in the case of Lane 1 where the vehicle density is 36, the signal duration is set at 33.54 seconds and in Lane 2 with a density of 25, the duration given is 23.29 seconds. In this case, Lane 3 with the biggest density of 60 is given time of 55.90 seconds, while for Lane 4 which has a density of 40 the descried time is 37.27 seconds. Fig. 5 demonstrates the dynamic adjustment of signal durations for each lane based on real-time traffic movements. These changes in light signals are made in accordance with the measured traffic values.

TABLE IV. COMPARISON WITH PREVIOUS IMPLEMENTATIONS

Related work	Task	Algorithms	Performance metrics
Deshpande and Hsieh [7]	Cyber-physical system for smart traffic light control	DarkNet53 for object detection SVM for adaptive traffic light control	Object detection accuracy: 97.5% Adaptive traffic light control accuracy: 98.19%
Bisht et al. [8]	Indigenous design of a traffic light control system responsive to the local traffic dynamics and priority vehicle	YOLO for vehicle detection Hybrid Genetic algorithm for adaptive traffic light control	Mean Average Precision:62.2% PCU error: 20%

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Periyasamy et al. [9]	Soft GRU-based RNN for enhanced congestion prediction for optimizing traffic flow in smart cities	Bi-directional RNN-GRU and LSTM for vehicle density estimation Random Forest algorithm for vehicle density estimation	Accuracy-87.5% Precision-94.36%
Saleem et al. [10]	Fusion-based intelligent traffic congestion control system for vehicular networks using machine learning techniques	FITCSS-VN for reducing traffic congestion by searching available routes SVM for adaptive traffic control	Accuracy: 95% Precision: 62.5.%
Alsaawy et al. [11]	A comprehensive and effective framework for traffic congestion problem based on the integration of IoT and data analytics	RFID for vehicle detection Porter's Algorithm Agents-Based Algorithm for adaptive traffic light control	Accuracy: 95%
Ali et al. [12]	An adaptive method for traffic signal control based on fuzzy logic with webster and modified webster formula using SUMO traffic simulator	Fuzzy logic algorithm for adaptive traffic light control	Accuracy: 90.32% IoU-67.5%
Neelakandan et al. [13]	Adaptive traffic signal control for intelligent transportation based on a consortium blockchain	OWENN algorithm for vehicle density estimation and traffic light control	Accuracy: 98.23% MAE-0.254 RMSE-0.345
Mathiane et al. [14]	A Vehicle Density Estimation Traffic Light Control System Using a Two-Dimensional Convolution Neural Network	2D-CNN for vehicle classification signal switching algorithm for adaptive traffic control.	Accuracy-96%
Yang et al. [15]	Multiclass objects detection algorithm using DarkNet-53 and DenseNet for intelligent vehicles	Darknet-53 for multi-class object detection DenseNet for vehicle detection	mAP-91.13% on KITTI dataset mAP-88.84% on Pascal VOC dataset
Reddy et al. [16]	Comparative Evaluation of Convolutional Neural Network Object Detection Algorithms for Vehicle Detection	FR-CNN for vehicle detection YOLO v3 for vehicle detection SSD for vehicle detection	FR-CNN: Accuracy:99.58% mAP: 76% YOLO v3: Accuracy: 99.47% mAP:81% SSD: Accuracy: 99.36% mAP:92%
Proposed work	Analysis of vehicle density and adaptive traffic light control using intelligent traffic management system.	FRCNN for vehicle detection GMM for density estimation Fuzzy logic algorithm for adaptive traffic light	Vehicle detection Accuracy: 93.2% Precision:95.4% Recall: 91% Vehicle Density MSE: 4.0% RMSE: 2.0% MAE: 2.0%

Table IV summarizes prior implementations relevant to the proposed work. Deshpande and Hsieh worked on Darknet53 for vehicle detection and SVM for adaptive traffic light control and achieved an accuracy of 97.5 % for vehicle detection and 98.19 % for adaptive traffic light control [7]. In addition, Bisht et al. trained the YOLO object detection model for vehicle detection and developed a Hybrid genetic algorithm for traffic signal control links following optimization with 62.2 % MAP and 20 % PCU error [8]. Abdullah et al. worked on a bi-directional RNN-GRU and LSTM for adaptive traffic control and random forest estimation for vehicle density. An example case study of adaptive traffic light control achieved 87.5 % accuracy and 94.36% precision [9]. Saleem et al. had applied FITCCS-VN for congestion reduction and SVM for adaptive traffic control which yielded 95% accuracy and 62.5% r2 error [10]. Alsaawy et al. application of Porter's algorithm agents-based Algorithm for Intelligent Control of Switched Traffic Lights achieved a 95% success rate [11]. Ali et al. employed fuzzy logic algorithm, which he used to control traffic lights and yielded an accuracy rate of 90.32% and 67.5% IoU [12]. Neelakandan et al. utilized OWENN algorithm for traffic density estimation and traffic signal control timing algorithms [13].

Mathiane et al. employed the 2D-CNN model for vehicle classification and signal switching algorithm in the design of adaptive traffic light control the rewards achieved accuracy of 96% [14]. Yang et al. proposed Multiclass object detecitn algorithm using DarkNet-53 and DenseNet achieving a mAP of 91.13% [15]. Reddy et al. explored three different algorithms: FR-CNN, which achieved 99.58% accuracy; YOLO v3, which achieved 99.47% accuracy; and SSD, which achieved 99.36% accuracy [16]. FR-CNN outperforms YOLO and SSD not only in accuracy but also due to its robustness in handling complex object detection scenarios and delivering higher precision in challenging environments. On the contrary, the proposed model provides an accuracy of 93.2%, a precision of 95.4%, and a recall of 91%. The FRCNN model performed significantly better than the other model architectures that were implemented due to its region proposing networks which were integrated with object detection systems with a good detection rate. Fuzzy logic as used in the proposed system gives a convenient and intelligent solution to traffic management by allowing convincing restrictions based on present road conditions. The combination of advanced detection and adaptive control algorithms in the proposed work enhances traffic management.

V. CONCLUSION

This article marks a significant advancement in the attempts to solve the current and ever-growing problem of urban traffic congestion. An extensive and rather active

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approach to handling traffic situations includes video processing, vehicle detection with a trained FRCNN model, and adaptive traffic light control using fuzzy logic techniques. This system's supremacy of adapting based on the real-time measurement of vehicle density presents urban traffic management in a new light as it provides an alternative to the normal timed controlled traffic lights that are smarter and reasonably dynamic. The system minimizes congestion hot spots and enhances the overall satisfaction of commuters by analyzing data and incorporating smart algorithms that enable the system to adjust to real-time traffic conditions. In this implementation, accuracy was an impressive 93.2%, precision 95.4% and recall 91%. These high figures demonstrate that the system is capable. Despite promising results, the system may struggle with real-time processing during peak traffic and is sensitive to extreme weather, affecting accuracy. Advanced machine learning techniques, like reinforcement learning, could be used in further research to improve predictive modeling and support non-vehicle traffic management, enabling dynamic adaptation.

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